

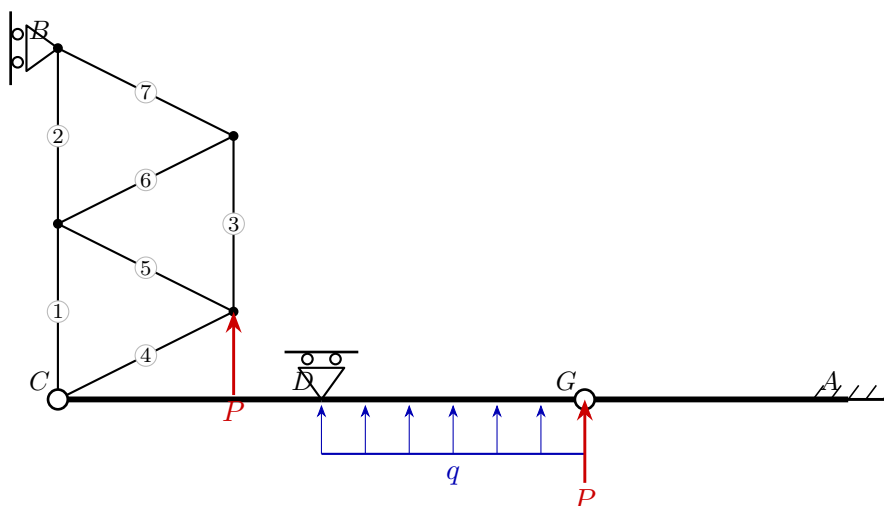
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam Group A

Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 3 - (5 + 4) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

 $A: R_x = -10, R_y = 2.5, M = -7.5 \quad D: R_x = 0, R_y = -47.5 \quad B: R_x = 10, R_y = 0$
 $C: H_x = 10, H_y = 20 \quad G: H_x = 10, H_y = -12.5$

3. Truss member forces

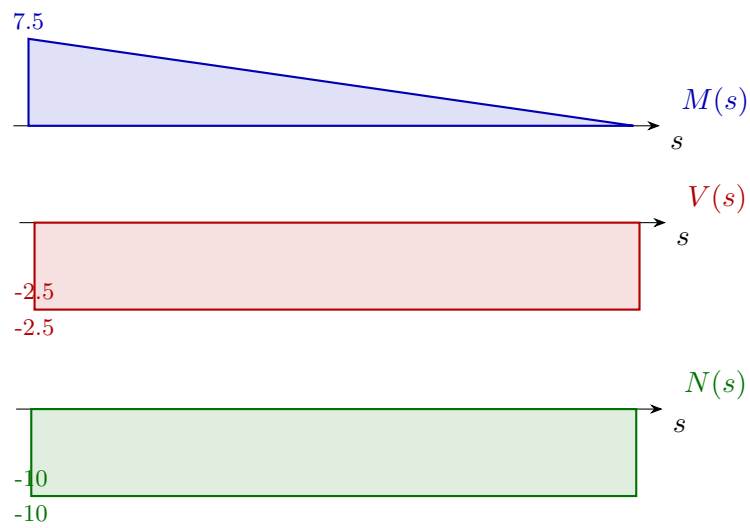
Member	Force S [kN]	Type
1	15	tension
2	5	tension
3	-10	compression
4	11.18	tension
5	-11.18	compression
6	11.18	tension
7	-11.18	compression

red: tension (+) blue: compression (–)



4. Section-force diagrams of the beams

Beam 1



Beam 2

